

NORTHERN UNIVERSITY OF BUSINESS AND TECHNOLOGY, KHULNA

Department of Computer Science & Engineering

AI SMART CAMPUS SYSTEM

Development Roadmap & Planning Document

Course	CSE4204 — Mobile Computing Lab
Team	CSE4204-8A-T07
Section	8A
Instructor	Md. Riaz Mahmud
Institution	NUBTK
Submission	June 2026

DEVELOPMENT ROADMAP

The development roadmap describes the planned activities for completing the AI Smart Campus System within the remaining weeks of the semester. The roadmap follows a logical development sequence starting from project setup and backend foundation, followed by frontend development, core feature implementation, AI integration, testing, documentation, and final submission.

The project will be developed using React.js for the frontend, Laravel REST API for the backend, MySQL for database management, and Google Gemini API for AI-based chatbot, recommendation, study plan generation, and risk prediction features.

Development Sequence Flow

The development sequence flow diagram illustrates the overall order of system development from project setup to final submission, covering all nine major phases of the project lifecycle.

Development Sequence Flow

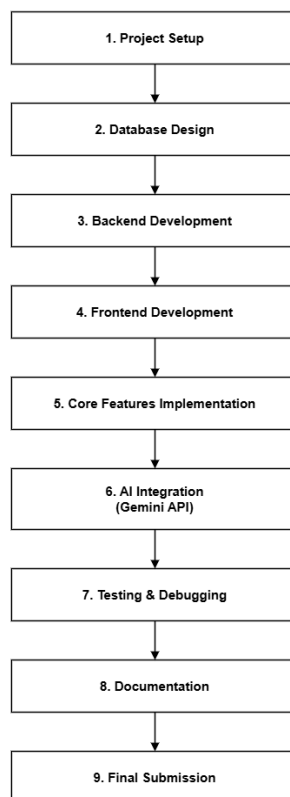


Figure 3.1: Development Sequence Flow

DEVELOPMENT ROADMAP TABLE

Week	Major Module	Major Tasks	Responsible Team Members	Expected Outcome
Week 06	Project Setup & Backend Foundation	• Create project structure • Configure MySQL database • Set up Laravel backend • Configure authentication system • Create Student, Faculty, and Admin models • Prepare role-based access structure	Backend Developer Database Manager	Backend environment ready, database connected, and authentication foundation completed.
Week 07	Frontend UI Development	• Develop landing page • Develop login page • Develop registration page • Create dashboard UI • Create navigation components • Implement responsive design	Frontend Developer UI/UX Designer	Complete user interface structure prepared with responsive layouts.
Week 08	Core Feature Development	• Student dashboard implementation • Faculty dashboard implementation • Admin dashboard implementation • Attendance module • Quiz result module • Mid-term performance module	Frontend Developer Backend Developer	Core system functionalities operational and connected with backend APIs.
Week 09	AI Integration	• Integrate Google Gemini API • Develop AI chat assistant • Develop course recommendation module • Generate personalized recommendations • Create study plan generation feature • Implement risk prediction logic	AI Integration Developer Backend Developer	AI-powered features functional and connected with academic performance data.
Week 10	Testing & Debugging	• Unit testing • Integration testing • API testing • Authentication testing • Role-based access testing • AI response testing • Fix UI issues and resolve bugs	QA Tester All Team Members	Stable, tested, and bug-reduced system ready for final improvement.

Week	Major Module	Major Tasks	Responsible Team Members	Expected Outcome
Week 11	Finalization & Documentation	- Final UI improvements - System optimization - Documentation update - GitHub repository cleanup - Final presentation preparation - Arrange final submission materials	Documentation Lead All Team Members	Complete project ready for demonstration and final submission.

Roadmap Explanation

The roadmap follows a logical sequence: backend and database work are completed first, frontend and core features are developed next, AI integration is added after the data flow is ready, and final testing and documentation are completed before submission.

Technology Stack Summary

Frontend	React.js
Backend	Laravel REST API
Database	MySQL
AI Integration	Google Gemini API (Chatbot, Recommendations, Study Plans, Risk Prediction)